

応用数学 第3章 §2 練習問題2 (p.104)

【フーリエ変換と積分定理】

問題 1

次の関数のフーリエ変換を求めよ.

$$(1) f(x) = \begin{cases} 2 & (0 \leq x < 2) \\ 1 & (2 \leq x < 3) \\ 0 & (x < 0, x \geq 3) \end{cases}$$

$$(2) g(x) = \begin{cases} 1 - x/2 & (0 \leq x < 2) \\ 0 & (x < 0, x \geq 2) \end{cases}$$

解答:

$$\begin{aligned} (1) F(u) &= \int_0^2 2e^{-iux} dx + \int_2^3 e^{-iux} dx \\ &= \frac{2(1 - e^{-2iu})}{iu} + \frac{e^{-2iu} - e^{-3iu}}{iu} = \frac{2 - e^{-2iu} - e^{-3iu}}{iu} \end{aligned}$$

$$(2) G(u) = \int_0^2 e^{-iux} dx - \frac{1}{2} \int_0^2 xe^{-iux} dx.$$

$$\text{第1項} = \frac{1 - e^{-2iu}}{iu}.$$

$$\text{第2項は部分積分で } \int_0^2 xe^{-iux} dx = -\frac{2e^{-2iu}}{iu} - \frac{1 - e^{-2iu}}{u^2}.$$

$$G(u) = \frac{1}{iu} + \frac{1 - e^{-2iu}}{2u^2}$$

問題 2

次の関数について, (1) はフーリエ正弦変換, (2) はフーリエ余弦変換を求めよ.

$$(1) f(x) = \begin{cases} \sin x & (|x| \leq \pi) \\ 0 & (|x| > \pi) \end{cases}$$

$$(2) g(x) = \begin{cases} \cos x & (|x| \leq \pi/2) \\ 0 & (|x| > \pi/2) \end{cases}$$

解答:

$$(1) f \text{ は奇関数. } F_s(u) = \int_0^\pi \sin x \sin ux dx$$

$$= \frac{1}{2} \int_0^\pi [\cos(1-u)x - \cos(1+u)x] dx$$

$$= \frac{1}{2} \left[\frac{\sin(1-u)\pi}{1-u} - \frac{\sin(1+u)\pi}{1+u} \right]$$

$$\sin(1 \mp u)\pi = \pm \sin u\pi \text{ より}$$

$$F_s(u) = \frac{\sin u\pi}{1-u^2} \quad (u \neq 1), \quad F_s(1) = \frac{\pi}{2}$$

$$(2) g \text{ は偶関数. } G_c(u) = \int_0^{\pi/2} \cos x \cos ux dx$$

$$= \frac{1}{2} \int_0^{\pi/2} [\cos(1-u)x + \cos(1+u)x] dx$$

$$= \frac{1}{2} \left[\frac{\sin(1-u)\pi/2}{1-u} + \frac{\sin(1+u)\pi/2}{1+u} \right]$$

$$\sin(1 \mp u)\pi/2 = \cos(u\pi/2) \text{ より}$$

$$G_c(u) = \frac{\cos(u\pi/2)}{1-u^2} \quad (u \neq 1)$$